

D56: Derivation of the Linear Scaling $N_{\text{comp}}(k) = k$ from Axioms C1–C4

Resolution of Open Problem OP-D41-1-A

Projective Dynamic Logo Framework

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Epistemic Status

This document resolves **Open Problem OP-D41-1-A**: the derivation of the linear scaling $N_{\text{comp}}(k) = k$ for the number of maximal partial closures on a k -particle– k -hole nuclear interface surface. The main result (Theorem 3.1) is an **unconditional theorem of axioms C1–C4**, proved via three lemmas that build on existing results of D16a [1], D29 [2], and D42 [3]. All steps are verified by the companion Python script `PDL_OP_D41_1_v1.py` deposited alongside this document. The connection $N_{\text{comp}}(k) = k \Rightarrow B(E2) \propto k$ (Open Problem OP-D41-1-B, Conjecture H_B [4]) requires the additional identification of the $E2$ operator within the PDL formalism and is explicitly **not** claimed here; it is identified as Open Problem OP-E2-PDL in Section 6.

Abstract

The Projective Dynamic Logo (PDL) framework derives nuclear structure from four combinatorial axioms on finite signed graphs [5, 6]. Document D41 [4] introduced the closure multiplicity conjecture H_B : for a nucleus at the boundary of an island of nuclear inversion dominated by a kp - kh intruder configuration, the number of maximal partial closures on the proton–neutron interface surface scales as $N_{\text{comp}}(k) = k$. The present document provides a complete proof of this scaling law from axioms C1–C4, without any free parameter. The proof rests on three lemmas: (L1) uniqueness of the maximal closure on a single interface unit (D16a); (L2) disjointness of the mixed-triangle sets of k distinct $(A) \wedge (B)$ couplings (D29); (L3) C3-irreducibility of k independent interface copies. Together these lemmas establish that $N_{\text{comp}}(k) = k$ exactly. Numerical verification is provided for $k = 1, \dots, 5$ via the companion Python script. The separate question of whether $B(E2) \propto N_{\text{comp}}(k)$ follows from C1–C4 (Open Problem OP-D41-1-B) is identified as requiring the formal identification of the $E2$ multipole operator in the PDL relational framework, and is left as an explicit open problem.

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1 Introduction and context

1.1 Motivation

Document D41 [4] confronted the PDL nuclear framework with the measurement by Ha et al. [7] of the reduced electric quadrupole transition probabilities $B(E2; 2_1^+ \rightarrow 0_1^+)$ in ^{84}Mo and ^{86}Mo . The ratio $B(E2)[^{84}\text{Mo}]/B(E2)[^{86}\text{Mo}] = 2.454$, together with the wave-function component count ratio $N_{\text{comp}}(^{86}\text{Mo})/N_{\text{comp}}(^{84}\text{Mo}) = 1.962$, were shown to be consistent with the PDL structural identity $T \approx (\Delta n + 1)^2 = 25$, where $T = R_{\text{surf}}(p)^2/R_{\text{sea}}(n) = 25.26$ is the nuclear binding unit and $\Delta n = n_d - n_u = 4$ is the valence-core asymmetry of the proton [8, 9]. The conjecture H_B was formulated in D41 but left without a derivation from axioms C1–C4; this derivation was identified as Open Problem OP-D41-1.

The companion exploratory documents D-exp-Zr [10] and the present analysis of the data of Escudeiro, Recchia, Lenzi et al. [11] on the mirror nuclei ^{47}Cr – ^{47}V and ^{49}Mn – ^{49}Cr extend the context in which $N_{\text{comp}}(k) = k$ is relevant. We note, however, that the present theorem is strictly confined to the island-of-inversion regime as defined in D41, and that its extension to mirror-nuclei anomalies requires a separate argument not given here.

1.2 Structure of this document

Section 2 recalls the PDL prerequisites in self-contained form, so that a reader familiar with nuclear spectroscopy but not with the PDL corpus can follow the proof. Section 3 states and proves the main theorem via three lemmas. Section 4 presents the numerical verification. Section 5 derives the corollary $R_{\text{surf}}(k) = kT$ and its relation to the conjecture H_B . Section 6 identifies the remaining open problem OP-E2-PDL. Section 7 concludes.

2 Prerequisites: PDL nuclear interface in self-contained form

This section is self-contained. A reader who has not read the prior PDL documents can verify every statement made here directly, either from the definitions given below or from the companion Python script.

2.1 The PDL proton and neutron

Within the PDL framework, the proton is the minimal hierarchical composite closure admissible under axioms C1–C4 on finite signed graphs. It is uniquely characterised by the integer quintuplet $(24, 28, 930, 10087, 11017)$, where $n_u = 24$ and $n_d = 28$ are the two valence-core sizes, $r_{\text{val}}(p) = 930$ is the stationary valence relational content, $R_{\text{sea}}(p) = 10087$ is the dynamical sea, and $R_{\text{tot}}(p) = 11017$ is the total relational budget [5, 12, 1, 13]. The neutron quintuplet is $(24, 28, 1032, 9960, 10992)$ [8]. The active surface of the proton is (D05 [14]):

$$R_{\text{surf}}(p) = 310 \varphi \approx 501.59, \quad \varphi = \frac{1+\sqrt{5}}{2}. \quad (1)$$

The nuclear binding unit is (D40 [9]):

$$T = \frac{R_{\text{surf}}(p)^2}{R_{\text{sea}}(n)} = \frac{(310 \varphi)^2}{9960} = 25.260 \approx (\Delta n + 1)^2 = 25, \quad (2)$$

with $\Delta n = n_d - n_u = 4$. The near-identity $T \approx (\Delta n + 1)^2$ holds to 1.04%; it is a structural fact, not an identity.

2.2 The minimal admissible closure K_4

Theorem 2.1 (D16a [1]). *No finite signed graph on $|V| \leq 3$ vertices satisfies all four axioms C1–C4 simultaneously. The complete signed graph on $|V| = 4$ vertices and $|E| = 6$ edges, denoted K_4 , is the unique minimal admissible closure. It admits exactly 8 coherent sign configurations, forming 4 pulsation pairs $\{s^{(1)}, s^{(2)}\} = \{s, -s\}$. No proper sub-closure of K_4 satisfies C1–C4.*

2.3 The $(A) \wedge (B)$ stability criterion

Definition 2.2 (Mixed triangle and cross-product, D29 [2]). A *mixed triangle* (e_i, e_j, p_k) contains two K_4 vertices $e_i, e_j \in V(K_4)$ and one proton vertex p_k . The *cross-product* at pulsation half-cycle $c \in \{1, 2\}$ is:

$$P_c = s_{\times}(e_i, p_k, c) \cdot s_{\times}(e_j, p_k, c) \in \{+1, -1\}, \quad (3)$$

where $s_{\times}(e, p, c)$ is the sign of the cross-edge (e, p) at half-cycle c .

Theorem 2.3 ($(A) \wedge (B)$ criterion, D29 [2]). *A mixed triangle (e_i, e_j, p_k) is stable (coherent at both half-cycles) if and only if:*

$$(A) : \quad P_1 = s_{K_4}^{(1)}(e_i, e_j) \quad (\text{initial alignment}), \quad (4)$$

$$(B) : \quad P_2 = -P_1 \quad (\text{phase-locking}). \quad (5)$$

The equivalence $(A) \wedge (B) \Leftrightarrow \text{stable}$ was verified exhaustively over all $8 \times 6 \times 16 = 768$ combinations of K_4 sign configurations, edges, and cross-signs, with zero counter-example [2].

Proposition 2.4 (Stable fraction, D29–D50 [2, 15]). *For any edge $(e_i, e_j) \in E(K_4)$, exactly 4 of the $2^4 = 16$ cross-sign configurations satisfy $(A) \wedge (B)$. The stable fraction per surface relation is $f_{\text{stable}} = 4/16 = 1/4$.*

2.4 Interface unit and the nuclear binding surface

Definition 2.5 (Interface unit, D41 [4]). An *interface unit* is one $(A) \wedge (B)$ -coupled proton–neutron pair crossing the shell gap G in a kp - kh nucleus. Its relational surface consists of all mixed triangles formed between one K_4 block and the proton active surface $R_{\text{surf}}(p)$. The effective surface size of one interface unit is $T = R_{\text{surf}}(p)^2 / R_{\text{sea}}(n)$ triangles.

Definition 2.6 (Maximal partial closure, D16a [1]). A *maximal partial closure* of an interface surface is a connected sub-configuration that satisfies all four axioms C1–C4 and is not a proper sub-configuration of any larger such sub-configuration.

3 Main theorem: $N_{\text{comp}}(k) = k$

Theorem (OP-D41-1-A resolved)

Theorem 3.1 ($N_{\text{comp}}(k) = k$, main result). *Let a kp - kh nucleus at the boundary of an island of nuclear inversion be modelled as k independent $(A) \wedge (B)$ -coupled proton–neutron interface units in the sense of Definition 2.5. The number of maximal partial closures (Definition 2.6) of the assembly satisfies:*

$$N_{\text{comp}}(k) = k. \quad (6)$$

This is an unconditional theorem of axioms C1–C4.

The proof proceeds through three lemmas.

3.1 Lemma L1: uniqueness of the maximal closure on one interface unit

Lemma 3.2 (L1 — Uniqueness). *The unique maximal C1+C2+C3+C4-admissible closure on a single interface unit of surface T is K_4 (up to the global S_4 symmetry). Therefore $N_{\text{comp}}(1) = 1$.*

Proof. By Theorem 2.1 (D16a), K_4 on $|V| = 4$ vertices is the unique minimal admissible closure; no configuration on $|V| \leq 3$ satisfies C1. Every proper sub-configuration of the interface unit has $|V| \leq 3$ vertices (since the interface unit is itself modelled on one K_4), and therefore fails C1. The interface unit itself satisfies all four axioms by construction. Hence K_4 is the only maximal C1+C2+C3+C4-admissible sub-configuration on a single interface unit: $N_{\text{comp}}(1) = 1$. $\square$

3.2 Lemma L2: disjointness of mixed-triangle sets

Lemma 3.3 (L2 — Disjointness). *Let $\mathcal{U}_1, \dots, \mathcal{U}_k$ be k distinct interface units, with K_4 blocks on vertex sets $V_1 = \{0, \dots, 3\}$, $V_2 = \{4, \dots, 7\}$, $\dots$, $V_k = \{4(k-1), \dots, 4k-1\}$, and let coupling i use cross-edge $(e_a^{(i)}, p_b^{(i)})$ for $e_a^{(i)} \in V_i$. The sets of mixed triangles associated with two distinct couplings $i \neq j$ are disjoint:*

$$\mathcal{T}(\mathcal{U}_i) \cap \mathcal{T}(\mathcal{U}_j) = \emptyset \quad \forall i \neq j. \quad (7)$$

Proof. A mixed triangle (e_a, e_b, p_c) in $\mathcal{T}(\mathcal{U}_i)$ requires $e_a, e_b \in V_i$ (two vertices from the i -th K_4 block) and p_c from the proton vertex set. A triangle in $\mathcal{T}(\mathcal{U}_j)$ requires $e'_a, e'_b \in V_j$ with $V_i \cap V_j = \emptyset$ for $i \neq j$ (the K_4 blocks are on disjoint vertex sets by construction). A single triangle cannot have two vertices in V_i and two vertices in V_j simultaneously, since a triangle has only three vertices. Therefore no triangle belongs to both $\mathcal{T}(\mathcal{U}_i)$ and $\mathcal{T}(\mathcal{U}_j)$. $\square$

Remark 3.4. Lemma L2 is a purely combinatorial statement about vertex disjointness. It uses no property of the signs and no property of the axioms beyond the construction of the vertex sets. It holds for any number $k \geq 2$ of interface units.

3.3 Lemma L3: C3-irreducibility of the assembly

Lemma 3.5 (L3 — C3-irreducibility). *Let $\mathcal{U}_1, \dots, \mathcal{U}_k$ be k interface units with no cross-edge between V_i and V_j for $i \neq j$. Then:*

- (a) *Each unit $\mathcal{U}_i$ is C3-connected within itself.*
- (b) *No connected component of the assembly spans two or more units.*
- (c) *Each $\mathcal{U}_i$ is C3-irreducible with respect to the other units.*

Proof. (a) Each $\mathcal{U}_i$ is modelled on K_4 , which is the complete graph on $|V_i| = 4$ vertices; it is connected by definition.

(b) With no cross-edge between V_i and V_j , the graph of the assembly is the disjoint union $\mathcal{U}_1 \sqcup \dots \sqcup \mathcal{U}_k$. In a disjoint union, the connected components are exactly the individual summands.

(c) C3 (axiom: the graph must be connected) holds within each $\mathcal{U}_i$ but not globally across units. Any sub-configuration spanning two units $\mathcal{U}_i \cup \mathcal{U}_j$ is disconnected and therefore fails C3. $\square$

3.4 Proof of the main theorem

Proof of Theorem 3.1. We must show that the k -unit assembly has exactly k maximal C1+C2+C3+C4-admissible sub-configurations.

By Lemma L3(b), the connected components of the assembly are exactly $\mathcal{U}_1, \dots, \mathcal{U}_k$. By axiom C3, any C1+C2+C3+C4-admissible sub-configuration must be connected, and hence must be entirely contained within one component $\mathcal{U}_i$.

By Lemma L1, the unique maximal admissible closure within $\mathcal{U}_i$ is $K_4^{(i)}$ itself: $N_{\text{comp}}(\mathcal{U}_i) = 1$ for each i .

Since no admissible closure can span two units (C3 fails), and each unit contributes exactly one maximal closure, the total count is:

$$N_{\text{comp}}(k) = \sum_{i=1}^k N_{\text{comp}}(\mathcal{U}_i) = \sum_{i=1}^k 1 = k. \quad \square \quad (8)$$

□

Remark 3.6 (Role of Lemma L2). Lemma L2 (disjointness of mixed-triangle sets) is not used in the proof of $N_{\text{comp}}(k) = k$ itself, which relies only on L1 and L3. Lemma L2 becomes essential in the companion sub-problem B investigation (Section 5 and Open Problem OP-E2-PDL): it establishes the independence of the k $(A) \wedge (B)$ couplings under the Indifference Lemma (D42 [3]), which is the key input to the incoherent-summation argument for $B(E2) \propto k$.

4 Numerical verification

The main theorem was verified by the companion Python script `PDL_OP_D41_1_v1.py`, which constructs the K_4 signed graph explicitly, enumerates all 64 sign assignments, identifies the 8 coherent configurations, and verifies $N_{\text{comp}}(k) = k$ for $k = 1, \dots, 5$ under the independence assumption. Key numerical outputs are reproduced in Table 1.

Table 1: Numerical verification of $N_{\text{comp}}(k) = k$ for $k = 1, \dots, 5$. The surface size kT is computed from PDL integers alone. The result $N_{\text{comp}}(k) = k$ holds exactly under the independence assumption (no cross-edges between units).

k	kT	$N_{\text{comp}}(k)$	$N_{\text{comp}}(k)/k$	Verified
1	25.260	1	1.000	✓
2	50.521	2	1.000	✓
3	75.781	3	1.000	✓
4	101.041	4	1.000	✓
5	126.302	5	1.000	✓

Additionally, the script verifies that the 8 coherent configurations of K_4 form 4 pulsation pairs (Theorem 2.1), and that the mixed-triangle sets of two distinct couplings $(0_A, 0_B)$ and $(1_A, 1_B)$ are disjoint (Lemma L2). The full output is deposited alongside this document on Zenodo.

5 Corollary and relation to conjecture H_B

Corollary 5.1 ($R_{\text{surf}}(k) = kT$). *Under the independence assumption of Theorem 3.1, the effective active surface of a kp - kh nucleus is:*

$$R_{\text{surf}}(k) = kT = k \frac{R_{\text{surf}}(p)^2}{R_{\text{sea}}(n)}. \quad (9)$$

Proof. Each of the k independent interface units carries an active surface of size T . By Theorem 3.1 and Lemma L2, the k units contribute independently (no shared mixed triangles). The total active surface is the sum of the individual contributions: $R_{\text{surf}}(k) = kT$. □

5.1 Relation to the conjecture H_B of D41

Conjecture H_B of D41 [4] states that for a kp - kh nucleus, $B(E2) \propto N_{\text{comp}}(k) = k$. From Theorem 3.1, we now know that $N_{\text{comp}}(k) = k$ is a theorem. The remaining question is whether $B(E2) \propto N_{\text{comp}}(k)$ follows from C1–C4.

By D32 (Proposition 3 [16]), the number of $(A) \wedge (B)$ -stable mixed triangles at fixed nuclear scale r is:

$$N_{\text{mix}}(k, r) = \frac{R_{\text{surf}}(k) K_0}{r^2} = \frac{kT K_0}{r^2} \propto k, \quad (10)$$

where K_0/r^2 cancels in any ratio between two nuclei at the same mass scale. Combining with Lemma L2 (disjointness of mixed-triangle sets from distinct couplings) and the Indifference Lemma (D42, Theorem [3]), which ensures that distinct relations contribute independently, the k interface units contribute *incoherently* to the transition matrix element M_{fi} :

$$|M_{fi}|^2 = \sum_{i=1}^k |m_i|^2 = k |m_1|^2 \propto k = N_{\text{comp}}(k), \quad (11)$$

provided that the $E2$ matrix element M_{fi} is identified with a sum over $(A) \wedge (B)$ -stable mixed triangles. This identification constitutes the formal gap described in Open Problem OP-E2-PDL below.

Numerically, $R_{\text{PDL}} = 2T/(\Delta n + 1)^2 = 2.021$, and the experimental ratio $B(E2)[^{84}\text{Mo}]/B(E2)[^{86}\text{Mo}] = 2.461 \pm 0.756$ lies 0.58σ from this prediction [4, 7]. The incoherent scenario ($B(E2) \propto k$, ratio = 2.000) is preferred at 0.61σ , while the coherent scenario ($B(E2) \propto k^2$, ratio = 4.000) is disfavoured at 2.04σ . This constitutes empirical support for the incoherent-summation argument, though not a proof.

Conjecture H_B (D41 [4])

For a nucleus at the boundary of an island of inversion dominated by a kp - kh intruder configuration, $B(E2) \propto N_{\text{comp}}(k) = k$, and therefore:

$$\frac{B(E2)[k_1p-k_1h]}{B(E2)[k_2p-k_2h]} = \frac{2T}{(\Delta n + 1)^2} \cdot \frac{k_1}{k_2}. \quad (12)$$

The factor $2T/(\Delta n + 1)^2 = 2.021$ differs from the exact ratio k_1/k_2 by 1.04%, which is the structural residual $T/(\Delta n + 1)^2 - 1$.

6 Open problem OP-E2-PDL

Open Problem OP-E2-PDL

Identify, within the PDL signed-graph formalism, the combinatorial object that corresponds to the nuclear electric quadrupole ($E2$) transition operator. Specifically: show from axioms C1–C4 that the $E2$ matrix element between two nuclear states is proportional to the number N_{mix} of $(A) \wedge (B)$ -stable mixed triangles connecting those states. Combined with Theorem 3.1 (this document) and D32 [16], this would elevate Conjecture H_B [4] to a theorem.

The difficulty lies in connecting the PDL relational structure (signed graphs, pulsating closures, mixed triangles) to the standard multipole expansion of nuclear electromagnetic transitions. The PDL formalism has already established the connection between N_{mix} and the Coulomb potential (D32, Prop. 3 [16]) and between R_{surf} and entropy (D37–D50 [17, 15]). The missing step is a direct identification of $T(E2)$ as a quadratic function of the mixed-triangle amplitudes.

7 Conclusion

We have proved from axioms C1–C4 that the number of maximal partial closures on a kp - kh nuclear interface surface scales exactly as $N_{\text{comp}}(k) = k$ (Theorem 3.1). The proof is complete in three steps: uniqueness of the maximal closure on a single interface unit (Lemma L1, building on D16a), disjointness of the mixed-triangle sets of distinct couplings (Lemma L2, building on D29), and C3-irreducibility of the k -unit assembly (Lemma L3, from axiom C3 directly). All steps are verified numerically for $k = 1, \dots, 5$.

This resolves Open Problem OP-D41-1-A. The companion sub-problem OP-D41-1-B — whether $B(E2) \propto N_{\text{comp}}(k)$ follows from C1–C4 — requires the formal identification of the $E2$ operator in the PDL framework (Open Problem OP-E2-PDL). The numerical evidence from Ha et al. [7] supports the proportionality at 0.61σ and disfavours the coherent-summation alternative at 2.04σ .

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A Epistemic status table

Table 2: Epistemic status of all claims in D56.

Claim	Status	Source
K_4 is unique minimal admissible closure	Theorem	D16a
8 coherent configs on K_4 , 4 pulsation pairs	Theorem	D01, D16a
$(A) \wedge (B)$: 192/768 stable, ratio = 1/4	Theorem	D29
$T = R_{\text{surf}}(p)^2 / R_{\text{sea}}(n) = 25.26$	Theorem	D05, D40
$T \approx (\Delta n + 1)^2 = 25$ to 1.04%	Structural fact	D40
Mixed-triangle sets of distinct couplings are disjoint (L2)	Theorem (this doc.)	D29 + algebra
C3-irreducibility of k independent units (L3)	Theorem (this doc.)	C3 axiom
$N_{\text{comp}}(1) = 1$ (L1)	Theorem	D16a
$N_{\text{comp}}(k) = k$ (main theorem)	Theorem	D56
$R_{\text{surf}}(k) = kT$ (corollary)	Theorem (corollary)	D56
$N_{\text{mix}}(k) \propto k$	Theorem	D32 Prop. 3
Incoherent summation: $ M_{fi} ^2 = k m_1 ^2$	Cand. theorem	D42 + L2
$B(E2) \propto N_{\text{comp}}(k)$	Conjecture H_B	D41
$E2$ operator identified in PDL formalism	Open OP-E2-PDL	Future work

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